

DNS Configuration Guide for Finalevolutiongroup.com

Overview

This document provides complete DNS configuration instructions for deploying Final Evolution Lab services under the `finalevolutiongroup.com` domain.

1. Required DNS Records

Core Records

Type	Name	Value	TTL	Purpose
A	@	<WEB_SERVER_IP>	300	Root domain → marketing site
CNAME	www	finalevolution-group.com	300	www redirect → root domain
CNAME	stream	<STREAM- ING_SERVER_HOST NAME>	300	Pixel Streaming signalling
CNAME	app	<WEB_APP_HOSTNA ME>	300	Web app (game client)
A	turn	<TURN_SERVER_IP >	300	TURN/STUN re- lay server
CNAME	api	<API_SERVER_HOS TNAME>	300	REST API end- points

Email Records (Optional)

Type	Name	Value	TTL	Purpose
MX	@	10 mail.finalevolutiongroup.com	3600	Email routing
TXT	@	v=spf1 include:_spf.google.com ~all	3600	SPF record
TXT	_dmarc	v=DMARC1; p=quarantine; rua=mailto:...	3600	DMARC policy

2. SSL/TLS Certificate Setup

Option A: Let's Encrypt (Free, Recommended)

```
# Install certbot
sudo apt update && sudo apt install -y certbot python3-certbot-nginx

# Obtain certificates for all subdomains
sudo certbot certonly --nginx \
  -d finalevolutiongroup.com \
  -d www.finalevolutiongroup.com \
  -d stream.finalevolutiongroup.com \
  -d app.finalevolutiongroup.com \
  -d api.finalevolutiongroup.com \
  -d turn.finalevolutiongroup.com

# Auto-renewal (add to crontab)
0 0 1 * * certbot renew --quiet
```

Option B: Cloudflare Origin Certificates

If using Cloudflare as your DNS provider:

1. Go to **SSL/TLS → Origin Server** in Cloudflare dashboard
2. Click **Create Certificate**
3. Select hostnames: `*.finalevolutiongroup.com` , `finalevolutiongroup.com`
4. Choose validity period (15 years recommended)
5. Download the certificate and private key
6. Install on your origin server:

```
sudo mkdir -p /etc/ssl/finalevolutiongroup
sudo cp origin-cert.pem /etc/ssl/finalevolutiongroup/cert.pem
sudo cp origin-key.pem /etc/ssl/finalevolutiongroup/key.pem
sudo chmod 600 /etc/ssl/finalevolutiongroup/key.pem
```

Option C: Wildcard Certificate with Let's Encrypt

```
sudo certbot certonly --manual --preferred-challenges dns \
  -d '*.finalevolutiongroup.com' \
  -d finalevolutiongroup.com
```

3. Cloudflare Configuration

Initial Setup

1. **Add Site:** Go to dash.cloudflare.com (https://dash.cloudflare.com) → Add Site → Enter `finalevolutiongroup.com`
2. **Select Plan:** Free plan is sufficient for initial deployment
3. **Update Nameservers:** At your domain registrar, change nameservers to:
 - `ns1.cloudflare.com` (actual values provided by Cloudflare)
 - `ns2.cloudflare.com`

DNS Records in Cloudflare

```
# Proxied (orange cloud) – gets Cloudflare CDN + DDoS protection
A      @      <WEB_SERVER_IP>      Proxied
CNAME  www    finalevolutiongroup.com  Proxied
CNAME  app    <WEB_APP_HOSTNAME>      Proxied
CNAME  api    <API_SERVER_HOSTNAME>    Proxied

# DNS-only (gray cloud) – for WebSocket/streaming (must NOT be proxied)
CNAME  stream <STREAMING_SERVER_HOSTNAME>  DNS Only
A      turn   <TURN_SERVER_IP>           DNS Only
```

IMPORTANT: `stream` and `turn` subdomains must be DNS-only (gray cloud). Cloudflare proxying interferes with WebSocket connections and TURN relay.

SSL/TLS Settings

1. **SSL Mode:** Full (Strict)
2. **Minimum TLS Version:** TLS 1.2
3. **Always Use HTTPS:** ON
4. **Automatic HTTPS Rewrites:** ON
5. **HSTS:** Enable with max-age 6 months

Security Settings

1. **WAF:** Enable managed rules
2. **Bot Fight Mode:** ON
3. **Under Attack Mode:** OFF (enable if under DDoS)
4. **Rate Limiting:** Create rule for `/api/*` (100 req/min per IP)

Page Rules

```
# Force HTTPS
http://*finalevolutiongroup.com/* → Always Use HTTPS

# Cache marketing site
finalevolutiongroup.com/* → Cache Level: Standard, Edge TTL: 4 hours

# No cache for streaming
stream.finalevolutiongroup.com/* → Cache Level: Bypass
```

4. Nginx Configuration

Marketing Website (finalevolutiongroup.com)

```
server {
    listen 443 ssl http2;
    server_name finalevolutiongroup.com www.finalevolutiongroup.com;

    ssl_certificate      /etc/ssl/finalevolutiongroup/cert.pem;
    ssl_certificate_key  /etc/ssl/finalevolutiongroup/key.pem;

    # NextJS static export or proxy to Next server
    root /var/www/finalevolutiongroup/out;
    index index.html;

    location / {
        try_files $uri $uri.html $uri/index.html =404;
    }

    # Security headers
    add_header X-Frame-Options "SAMEORIGIN" always;
    add_header X-Content-Type-Options "nosniff" always;
    add_header X-XSS-Protection "1; mode=block" always;
    add_header Referrer-Policy "strict-origin-when-cross-origin" always;
    add_header Content-Security-Policy "default-src 'self'; script-src 'self' 'unsafe-
inline'; style-src 'self' 'unsafe-inline' fonts.googleapis.com; font-src fonts.gstatic
.com;" always;

    # Gzip
    gzip on;
    gzip_types text/plain text/css application/json application/javascript text/xml;
}

server {
    listen 80;
    server_name finalevolutiongroup.com www.finalevolutiongroup.com;
    return 301 https://$host$request_uri;
}
```

Streaming Signalling Server (stream.finalevolutiongroup.com)

```
server {
    listen 443 ssl http2;
    server_name stream.finalevolutiongroup.com;

    ssl_certificate      /etc/ssl/finalevolutiongroup/cert.pem;
    ssl_certificate_key  /etc/ssl/finalevolutiongroup/key.pem;

    location / {
        proxy_pass http://127.0.0.1:8888;
        proxy_http_version 1.1;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection "upgrade";
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_read_timeout 86400;
    }
}
```

Web App (app.finalevolutiongroup.com)

```
server {
    listen 443 ssl http2;
    server_name app.finalevolutiongroup.com;

    ssl_certificate      /etc/ssl/finalevolutiongroup/cert.pem;
    ssl_certificate_key  /etc/ssl/finalevolutiongroup/key.pem;

    root /var/www/fel-webapp/dist;
    index index.html;

    location / {
        try_files $uri $uri/ /index.html;
    }
}
```

5. Service Configuration Updates

After DNS is configured, update these files:

iOS App Config

```
// ios/FinalEvolutionLab/Config.swift
static let STREAMING_SERVER_URL = "wss://stream.finalevolutiongroup.com"
static let STREAMING_API_URL = "https://stream.finalevolutiongroup.com"
static let TURN_SERVER_URL = "turn:turn.finalevolutiongroup.com:3478"
static let STUN_SERVER_URL = "stun:turn.finalevolutiongroup.com:3478"
static let WEBSITE_URL = "https://finalevolutiongroup.com"
static let WEB_APP_URL = "https://app.finalevolutiongroup.com"
```

UE5 Pixel Streaming Config

```
# FEL_PixelStreaming_iOS.ini
SignallingServerURL=wss://stream.finalevolutiongroup.com
```

Docker Compose Environment

```
environment:
  - SIGNALLING_URL=wss://stream.finalevolutiongroup.com
  - TURN_SERVER=turn:turn.finalevolutiongroup.com:3478
  - PUBLIC_IP=<SERVER_PUBLIC_IP>
```

6. Verification Checklist

```
# Verify DNS propagation
dig finalevolutiongroup.com +short
dig www.finalevolutiongroup.com +short
dig stream.finalevolutiongroup.com +short
dig turn.finalevolutiongroup.com +short
dig app.finalevolutiongroup.com +short

# Verify SSL certificates
openssl s_client -connect finalevolutiongroup.com:443 -servername finalevolution-
group.com </dev/null 2>/dev/null | openssl x509 -noout -dates

# Verify website
curl -sI https://finalevolutiongroup.com | head -5

# Verify streaming WebSocket
wscat -c wss://stream.finalevolutiongroup.com

# Verify TURN server
turnutils_uclient -T turn.finalevolutiongroup.com
```

7. Troubleshooting

Issue	Solution
DNS not resolving	Wait 24-48h for propagation; check with <code>dig @8.8.8.8</code>
SSL error	Verify cert covers subdomain; check Cloudflare SSL mode
WebSocket 502	Ensure stream subdomain is DNS-only (not proxied)
TURN connection failed	Open UDP ports 3478, 49152-65535 on fire-wall
www not redirecting	Add CNAME for www pointing to root domain